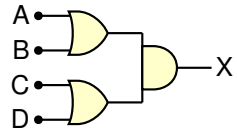
Homework 4, COSC330/PHYS306
Chapters 5 and 6

Prof. Jordan C. Hanson

1 Chapter 5

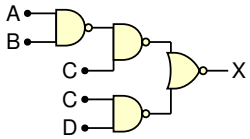
1. **[Section 5-1, Exercise 6]**
   Develop the truth table for each circuit in Figure 5–56. tables: p. 3-4

$X = (A+B)(C+D)$



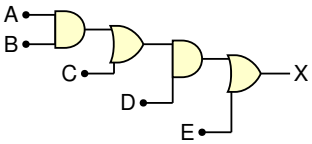
(a)

$X = \overline{AB} \cdot CD$



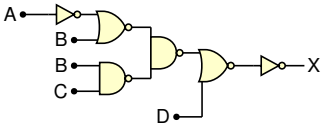
(b)

$X = (AB+C)D + E$



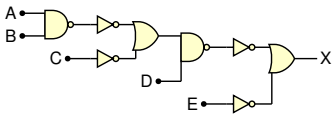
(c)

$X = \overline{A} + B + D$



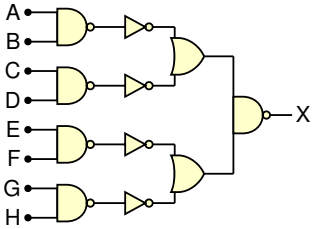
(d)

$X = (AB+\overline{C})D + \overline{E}$



(e)

$X = \overline{(AB+CD)(EF+GH)}$



(f)

FIGURE 5–56

2. **[Section 5-2, Exercise 13]**
   Implement a logic circuit for the truth table in Table 5–8.

$X = \overline{C} + AB$

Inputs			Output
A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

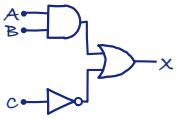
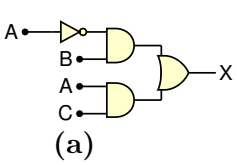
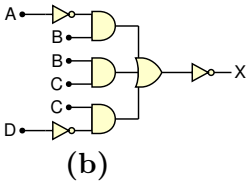


TABLE 5–8

3. **[Section 5-3, Exercise 20]**
   Implement the logic circuits in Figure 5–54 using only NAND gates.



(a)



(b)

FIGURE 5–54

NAND: p. 4

2 Chapter 6

4. **[Section 6-2, Exercise 4]**
   For the parallel adder in Figure 6–69, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two input numbers.

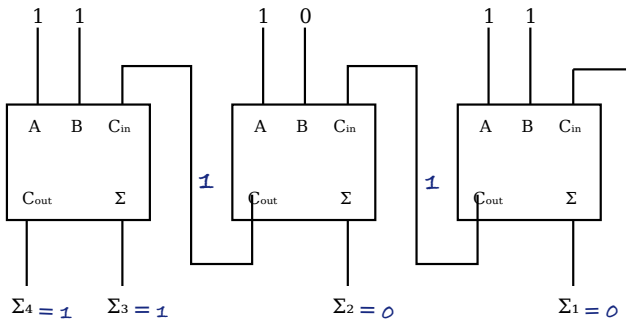


FIGURE 6–69 Sum = 1100

111 + 101 = 1100 (7 + 5 = 12) ✓

5. **[Section 6-2, Exercise 8]**
The input waveforms in Figure 6–72 are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.

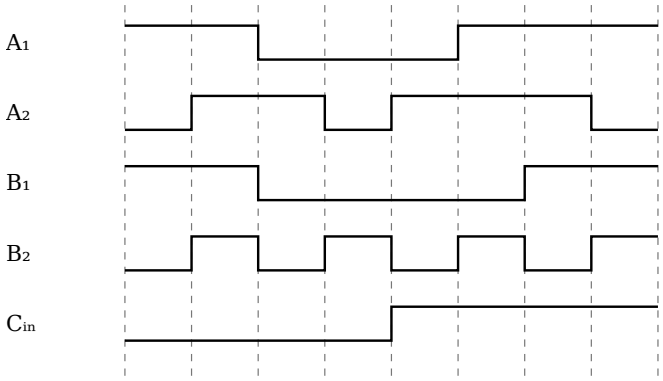


FIGURE 6–72 (see right →)

6. **[Section 6-4, Exercise 14]**
For the 4-bit comparator in Figure 6–74, plot each output waveform for the inputs shown. The outputs are active-HIGH.

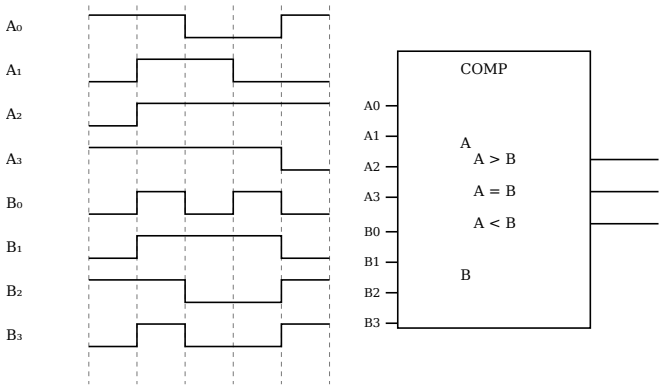


FIGURE 6–74

7. **[Section 6-5, Exercise 16]**
When a HIGH is on the output of each of the decoding gates in Figure 6–75, what is the binary code appearing on the inputs? The MSB is A_3.

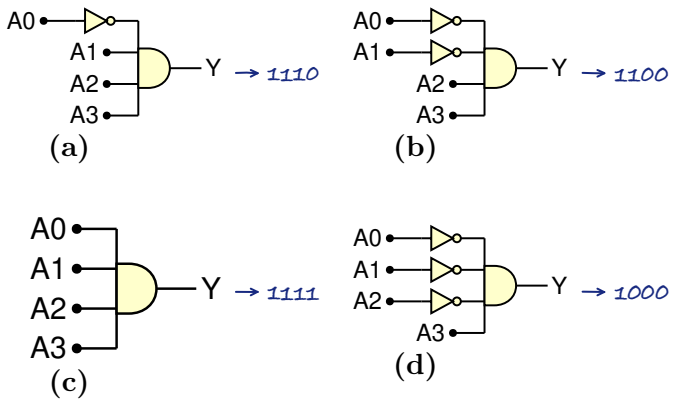


FIGURE 6–75

8. **[Section 6-8, Exercise 28]**
For the multiplexer in Figure 6–79, determine the output for the following input states:

$$D_0 = 0, \quad D_1 = 1, \quad D_2 = 1, \quad D_3 = 0,$$

$$S_0 = 1, \quad S_1 = 0.$$

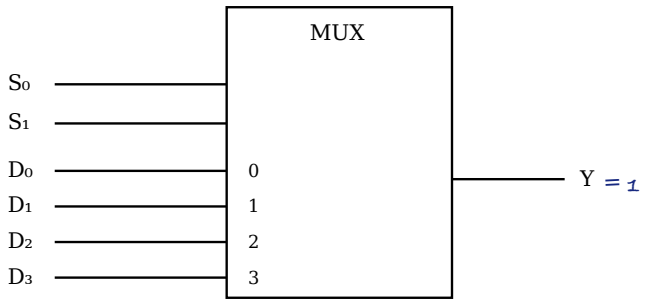
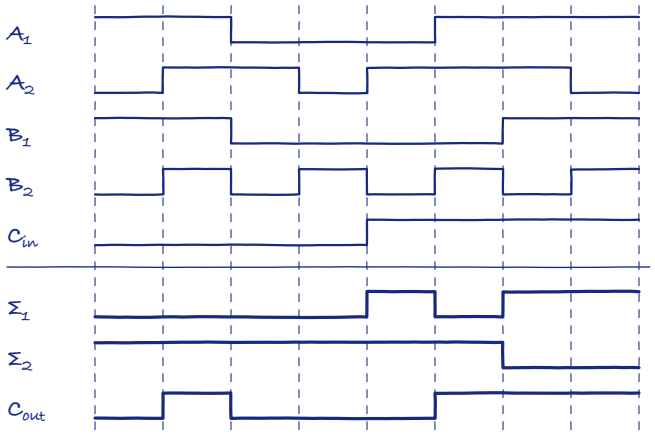


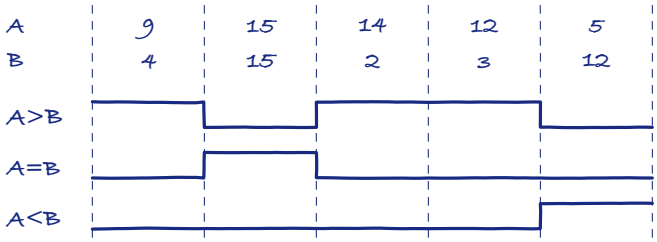
FIGURE 6–79

8. $S_1S_0 = 01 \rightarrow Y = D_1 = 1$

5.



6.



1. Truth tables (Figure 5-56)

(a) $X = (A+B)(C+D)$

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

(b) $X = \overline{AB} \cdot CD$

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

(d) $X = \bar{A} + B + D$

A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

(c) $X = (AB+C)D + E$

A	B	C	D	E	X
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	0
0	0	0	1	1	1
0	0	1	0	0	0
0	0	1	0	1	1
0	0	1	1	0	1
0	0	1	1	1	1
0	1	0	0	0	0
0	1	0	0	1	1
0	1	0	1	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	0	1	1
0	1	1	1	0	1
0	1	1	1	1	1

A	B	C	D	E	X
1	0	0	0	0	0
1	0	0	0	1	1
1	0	0	1	0	0
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	0	1	1
1	0	1	1	0	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	0	1	1
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	0
1	1	1	0	1	1
1	1	1	1	0	1
1	1	1	1	1	1

(e) $X = (AB+\bar{C})D + \bar{E}$

A	B	C	D	E	X
0	0	0	0	0	1
0	0	0	0	1	0
0	0	0	1	0	1
0	0	0	1	1	1
0	0	1	0	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	0	1	1	1	0
0	1	0	0	0	1
0	1	0	0	1	0
0	1	0	1	0	1
0	1	0	1	1	1
0	1	1	0	0	1
0	1	1	0	1	0
0	1	1	1	0	1
0	1	1	1	1	0

A	B	C	D	E	X
1	0	0	0	0	1
1	0	0	0	1	0
1	0	0	1	0	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	0	1	0
1	0	1	1	0	1
1	0	1	1	1	0
1	1	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	0	1	0
1	1	1	1	0	1
1	1	1	1	1	1

1. (f) $X = \overline{(AB+CD)(EF+GH)}$

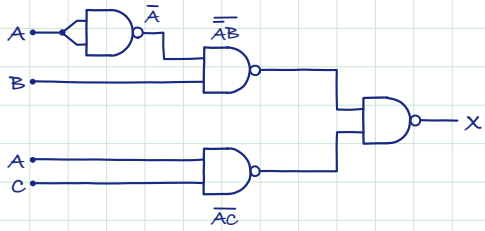
ABCD \ EFGH →	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0000	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0001	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0010	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0011	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
0100	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0101	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0110	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0111	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1000	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1001	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1010	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1011	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1100	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1101	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1110	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0
1111	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0	0

All 256 rows: row = ABCD, column = EFGH, entry = X

$X = 0$ only when
$AB+CD = 1$ and $EF+GH = 1$
(49 of 256 rows)

3. NAND only (Figure 5-54)

(a) $X = \bar{A}B + AC = \overline{\overline{\bar{A}B} \cdot \overline{AC}}$



(b) $X = \overline{\bar{A}B + BC + C\bar{D}}$

